

CSA05 – Database Management Systems

Assessment Tool 2 – Case Study

1. Online Shopping Platform – Normalize to 3NF

Relation:

Order(OrderID, CustID, CustName, ProdID, ProdName, Qty, Price)

FDs:

- OrderID $\rightarrow$ CustID
- CustID $\rightarrow$ CustName
- ProdID $\rightarrow$ ProdName, Price
- (OrderID, ProdID) $\rightarrow$ Qty

3NF Decomposition:

- CUSTOMER(CustID, CustName)
- PRODUCT(ProdID, ProdName, Price)
- ORDERS(OrderID, CustID)
- ORDER_ITEM(OrderID, ProdID, Qty)

Result: Redundancy and transitive dependencies are removed, achieving 3NF.

2. Hospital – Anomalies and BCNF

Relation:

Patient(PID, PName, DocID, DocName, Specialization, WardNo, WardName)

FDs:

- PID $\rightarrow$ PName, DocID, WardNo
- DocID $\rightarrow$ DocName, Specialization
- WardNo $\rightarrow$ WardName

Anomalies: Update, insertion, and deletion anomalies.

BCNF Decomposition:

- PATIENT(PID, PName, DocID, WardNo)
- DOCTOR(DocID, DocName, Specialization)
- WARD(WardNo, WardName)

Result: Every determinant is a key in its relation; therefore, the schema is in BCNF.

3. University Enrollment – 2NF

Relation:

Enrollment(SID, SName, CourseID, CourseName, Faculty, Grade)

FDs:

- $SID \rightarrow SName$
- $CourseID \rightarrow CourseName, Faculty$
- $(SID, CourseID) \rightarrow Grade$

Candidate Key: (SID, CourseID)

Partial Dependencies:

- $SID \rightarrow SName$
- $CourseID \rightarrow CourseName, Faculty$

2NF Decomposition:

- STUDENT(SID, SName)
- COURSE(CourseID, CourseName, Faculty)
- ENROLLMENT(SID, CourseID, Grade)

Result: Partial dependencies are removed and the relations are in 2NF.

4. Airline Booking – Normalize to BCNF

Relation:

Booking(FlightNo, PassID, PassName, Seat, DeptCity, ArrCity, Date)

FDs:

- $PassID \rightarrow PassName$
- $FlightNo \rightarrow DeptCity, ArrCity$
- $(FlightNo, PassID, Date) \rightarrow Seat$

BCNF Decomposition:

- PASSENGER(PassID, PassName)
- FLIGHT(FlightNo, DeptCity, ArrCity)

- BOOKING(FlightNo, PassID, Date, Seat)

Result: All determinants are candidate keys of their respective relations; hence the schema is in BCNF.

5. HR System – Normalize to 3NF

Assume relation:

EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID, ManagerName)

FDs:

- EmpID $\rightarrow$ EmpName, DeptID
- DeptID $\rightarrow$ DeptName, ManagerID
- ManagerID $\rightarrow$ ManagerName

Transitive Dependencies:

- EmpID $\rightarrow$ DeptID $\rightarrow$ DeptName, ManagerID
- EmpID $\rightarrow$ ManagerID $\rightarrow$ ManagerName

3NF Decomposition:

- EMPLOYEE(EmpID, EmpName, DeptID)
- DEPARTMENT(DeptID, DeptName, ManagerID)
- MANAGER(ManagerID, ManagerName)

Result: Transitive dependencies are eliminated, achieving 3NF.

6. Library – MVDs and 4NF

Relation:

Library(MemberID, MemberName, BookID, Title, Author, Genre, BorrowDate)

Relevant dependencies:

- MemberID $\rightarrow$ MemberName
- BookID $\rightarrow$ Title, Author, Genre
- MemberID $\twoheadrightarrow$ BookID (*independent multiple borrowed books*)

4NF Decomposition:

- MEMBER(MemberID, MemberName)
- BOOK(BookID, Title, Author, Genre)
- BORROW(MemberID, BookID, BorrowDate)

Result: Multivalued information is separated, reducing unnecessary combinations and achieving 4NF.

7. Telecom Billing – Keys and Functional Dependencies

Assume:

BILLING(CustomerID, CustomerName, MobileNo, BillNo, BillDate, Amount)

Functional Dependencies:

- CustomerID $\rightarrow$ CustomerName, MobileNo
- BillNo $\rightarrow$ CustomerID, BillDate, Amount

Candidate Keys:

- BillNo

If MobileNo is unique:

- MobileNo, BillNo identifies a specific billing record but is not minimal if BillNo alone identifies it.

Primary Key: BillNo

Result: Functional dependencies and keys are identified before normalization.

8. Retail Inventory – Complete Normalization

Relation:

Product(PID, PName, SupplierID, SupplierName, Category, Price, Warehouse)

FDs:

- PID $\rightarrow$ PName, SupplierID, Category, Price, Warehouse
- SupplierID $\rightarrow$ SupplierName

Transitive Dependency:

- $PID \rightarrow SupplierID \rightarrow SupplierName$

Normalized Relations:

- `PRODUCT(PID, PName, SupplierID, Category, Price, Warehouse)`
- `SUPPLIER(SupplierID, SupplierName)`

Result: The transitive dependency is removed. The resulting schema satisfies 3NF/BCNF, assuming PID and SupplierID are keys of their respective relations.

9. School ERP – Normalize to BCNF

Assume poorly designed relation:

`SCHOOL(StudentID, StudentName, ClassID, ClassName, TeacherID, TeacherName, SubjectID, SubjectName)`

FDs:

- $StudentID \rightarrow StudentName, ClassID$
- $ClassID \rightarrow ClassName, TeacherID$
- $TeacherID \rightarrow TeacherName$
- $SubjectID \rightarrow SubjectName$

BCNF Design:

- `STUDENT(StudentID, StudentName, ClassID)`
- `CLASS(ClassID, ClassName, TeacherID)`
- `TEACHER(TeacherID, TeacherName)`
- `SUBJECT(SubjectID, SubjectName)`
- `CLASS_SUBJECT(ClassID, SubjectID)`

Justification: Each determinant is a key in its corresponding relation, removing redundancy and anomalies.

10. Movie Streaming Platform – 5NF

Assume relation:

`STREAMING(UserID, MovieID, PlatformID)`

Possible join dependency:

$\text{STREAMING} = \text{USER_MOVIE} \bowtie \text{MOVIE_PLATFORM} \bowtie \text{USER_PLATFORM}$

Decompose into:

- $\text{USER_MOVIE}(\text{UserID}, \text{MovieID})$
- $\text{MOVIE_PLATFORM}(\text{MovieID}, \text{PlatformID})$
- $\text{USER_PLATFORM}(\text{UserID}, \text{PlatformID})$

Lossless-Join Verification:

Joining the three decomposed relations reconstructs the valid tuples of the original relation without information loss:

$\text{USER_MOVIE} \bowtie \text{MOVIE_PLATFORM} \bowtie \text{USER_PLATFORM} = \text{STREAMING}$

Result: Redundancy caused by join dependencies is removed, and the schema is decomposed to 5NF (PJNF).